

$$1. A) \alpha = P(|Y - 18| \geq 4 | p = 0.5) = P(Y \leq 14) + P(Y \geq 22)$$

The R code below finds the probabilities $P(Y \leq 14)$ and $P(Y \geq 22)$.

```
binom.test(14,36,0.5,alternative="less")

##
## Exact binomial test
##
## data: 14 and 36
## number of successes = 14, number of trials = 36, p-value = 0.1215
## alternative hypothesis: true probability of success is less than 0.5
## 95 percent confidence interval:
##  0.000000 0.539511
## sample estimates:
## probability of success
##           0.3888889

binom.test(22,36,0.5,alternative="greater")

##
## Exact binomial test
##
## data: 22 and 36
## number of successes = 22, number of trials = 36, p-value = 0.1215
## alternative hypothesis: true probability of success is greater than 0.5
## 95 percent confidence interval:
##  0.460489 1.000000
## sample estimates:
## probability of success
##           0.6111111
```

$$\text{Then } \alpha = 0.1215 + 0.1215 = 0.243$$

$$B) \beta = P(|Y - 18| \geq 3 | p = 0.7) = P(15 \leq Y \leq 21)$$

The R code below finds the probabilities $P(Y \leq 15)$ and $P(Y \leq 21)$.

```
binom.test(15,36,0.7,alternative="less")

##
## Exact binomial test
##
## data: 15 and 36
## number of successes = 15, number of trials = 36, p-value = 0.0003845
## alternative hypothesis: true probability of success is less than 0.7
## 95 percent confidence interval:
##  0.0000000 0.5669144
## sample estimates:
```

```
## probability of success
##          0.4166667

binom.test(21,36,0.7,alternative="less")

##
## Exact binomial test
##
## data:  21 and 36
## number of successes = 21, number of trials = 36, p-value = 0.09166
## alternative hypothesis: true probability of success is less than 0.7
## 95 percent confidence interval:
##  0.000000 0.722805
## sample estimates:
## probability of success
##          0.5833333
```

Then $\beta = 0.09166 - 0.0003845 = 0.0912755$

2. $\beta(\mu_a) = 1 - 2\Phi\left(\frac{\mu_0 - \mu_a}{s/\sqrt{n}} + z_{\alpha/2}\right)$ where $z_{\alpha/2}$ is the upper alpha percentile

$$\frac{1 - \beta(\mu_a)}{2} = \Phi\left(\frac{\mu_0 - \mu_a}{s/\sqrt{n}} + z_{\alpha/2}\right)$$

$$\Phi^{-1}\left(\frac{1 - \beta(\mu_a)}{2}\right) = \frac{\mu_0 - \mu_a}{s/\sqrt{n}} + z_{\alpha/2}$$

$$z_{\frac{1 - \beta(\mu_a)}{2}} = \frac{\mu_0 - \mu_a}{s/\sqrt{n}} + z_{\alpha/2}$$

$$z_{\frac{1 - \beta(\mu_a)}{2}} - z_{\alpha/2} = \frac{\mu_0 - \mu_a}{s/\sqrt{n}}$$

$$\frac{z_{\frac{1 - \beta(\mu_a)}{2}} - z_{\alpha/2}}{\frac{\mu_0 - \mu_a}{s}} = \frac{\sqrt{n}}{s}$$

$$\frac{s(z_{\frac{1 - \beta(\mu_a)}{2}} - z_{\alpha/2})}{\mu_0 - \mu_a} = \sqrt{n}$$

$$\left[\left(\frac{s(z_{\frac{1 - \beta(\mu_a)}{2}} - z_{\alpha/2})}{\mu_0 - \mu_a} \right)^2 \right] = n$$

3. A) $H_0: \mu = 130$ $H_a: \mu \neq 130$

$$TS = \frac{128.6 - 130}{2.1/\sqrt{40}} = -4.2164$$

$$RR = |TS| > z_{0.975} \quad RR = |-4.2164| > 1.96$$

The R code below finds the p-value for $z = -4.2164$

```
pnorm(-4.2164,lower.tail=TRUE)
```

```
## [1] 1.241166e-05
```

With a very small p-value ~ 0 , we reject the null hypothesis. We have very strong evidence that the true output voltage mean is significantly different from 130.

$$\begin{aligned} \text{B) } \beta(128.6) &= 1 - 2\Phi\left(\frac{128.6 - 128.6}{2.1/\sqrt{40}} - 1.96\right) \\ &= 1 - 2\Phi(-1.96) \end{aligned}$$

$$\beta(128.6) = 1 - 2(0.025) = 1 - 0.05 = 0.95$$

$$\text{C) } n = \left\lceil \left(\frac{2.1(z_{1-0.91} - z_{0.025})}{128.6 - 130} \right)^2 \right\rceil$$

The R code below finds the quantile values for $z_{0.045}$ and $z_{0.025}$

```
qnorm(0.045,lower.tail=TRUE)
```

```
## [1] -1.695398
```

```
qnorm(0.025,lower.tail=TRUE)
```

```
## [1] -1.959964
```

$$n = \left\lceil \left(\frac{2.1(-1.695398 - 1.959964)}{-1.4} \right)^2 \right\rceil$$

$$n = \left\lceil \left(\frac{2.1(-3.655362)}{-1.4} \right)^2 \right\rceil$$

$$n = \lceil (5.483043)^2 \rceil$$

$$n = \lceil 30.06376054 \rceil$$

Then the minimum sample size needed for the power to be 0.91 is 30.06376054.

$$4. \text{ A) } H_0: p_{0.6} \leq p_{0.7} \quad H_0: p_{0.6} > p_{0.7}$$

$$TS = \frac{\frac{55}{70} - \frac{23}{70}}{\sqrt{\frac{78}{140}\left(1 - \frac{78}{140}\right)\left(\frac{1}{70} + \frac{1}{70}\right)}}$$

$$= \frac{0.786 - 0.329}{\sqrt{(0.557)(0.443)(\frac{2}{70})}} = 5.443$$

The R code below calculates the p-value for $z=5.443$

```
pnorm(5.443, lower.tail=FALSE)
```

```
## [1] 2.619532e-08
```

B) With a very small p-value ~ 0 , we reject the null hypothesis. We have very strong evidence that the true rate of normal RNA synthesis is lower for cells exposed to the higher concentrations of actinomycin D.